

Mayur Joshi

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Profile Summary

- Data Scientist with **6+** years of experience in developing end-to-end data science products and delivering client projects.
- Strong expertise in data pre-processing, machine learning algorithms, and deep learning model training and deployment.
- Experience with advanced **NLP** techniques, including Question-Answering, Information Retrieval, Sentiment Analysis, and Clustering, utilizing **LLMs** such as **GPT-3.5/4**, **Gemini**, and **LlaMA**.
- Proficient in implementing Retrieval-Augmented Generation (**RAG**) approaches to enhance information retrieval tasks.
- Hands-on experience with NLP tools and **embeddings**, including **BERT**, **USE**, and **Word2Vec**.
- Experienced in computer vision models, transfer learning, and deployments using **Yolov5**, **ResNet**, and **VGG**.
- Skilled in leveraging **Hugging Face** libraries and tools to fine-tune and deploy large language models.
- Proficient in Python libraries such as **Sklearn**, **Numpy**, **Pandas**, **Matplotlib**, **TensorFlow**, and **Keras**.

SKILLS

- Python, SQL, Shell Scripting
- Machine Learning: - **KNN**, **SVM**, **Xgboost**, **Random Forest**, **Decision Tree**, **Logistic Regression**.
- Deep Learning: - **LSTM**, **GRU**, **CNN**, **MLP**
- LLMs :- **Gemini**, **GPT-3.5/4**, **LlaMA**
- Time Series Forecasting: - **S-ARIMA**, **Holt-Winters**, **Prophet**
- **Langchain**, Scikit Learn, TensorFlow, Pytorch, Keras
- AWS Services, GCP
- Apache Solr, Elasticsearch
- Django, Flask, BentoML
- **Docker**, Kubernetes, CI/CD
- AirFlow

WORK EXPERIENCE

Lead Data Scientist

Vizerto by DigitalMain, Pune | June 2019 – April 2022; May 2023 To Present

Gen-AI Based Information Retrieval System (RAG):-

Developed a GenAI-powered digital assistant and knowledge amplifier for B2B sellers, providing credible and rapid answers to customer, business, and technical questions.

Roles and Responsibilities: -

- Set up **Solr** core for full-text search across question-answer pairs, PDF content, web crawls, collaboration channels, and **vector data**, optimizing overall search infrastructure.
- Enhanced **Solr** search functionality through **multiword synonyms** and **Edismax** query parsing for improved domain-specific accuracy.
- Developed a machine learning-based relevancy classifier using word share, distance metrics, fuzzy logic, and vector techniques to enhance search result ranking.
- Implemented domain-specific **query rewriting** using **LLMs** and historical questions to improve user query understanding, effectively handling comparison and descriptive queries to provide detailed responses.
- Utilized **embedding**-based retrieval and **cross-encoder** re-ranking models, fine-tuning **BERT** models (MLM, NLI) to achieve **89%** accuracy on top search results.
- Evaluated various retrieval and re-ranking models (**BERT**, **USE**, **Word2Vec**) to optimize search accuracy.
- Developed generative AI capabilities for real-time answer generation leveraging **GPT-3.5/4** and **Google Gemini models**, utilizing in-context retrieved data to enhance response quality.
- Implemented **zero-shot** and **few-shot prompt** fine-tuning to minimize hallucinations and improve contextual adherence.
- Created metrics to evaluate generative AI responses, focusing on **context adherence and correctness**.
- Handled the software development life cycle from scratch using an Agile sprint cycle to ensure timely delivery of project milestones.
- Designed **REST APIs** for all services using **Flask** and deployed code using **Gunicorn**, **Nginx**, **Docker**, **CI/CD** pipelines on **AWS** for scalable application management.

Automated Data Extraction and Integration for Collaboration Platforms

Developed a weekly automated data scraping process for platforms such as **Slack**, **Microsoft Teams**, **Webex**, and **Jira**, **Gmail** and **Outlook** streamlining information retrieval workflows.

Roles and Responsibilities: -

- Leveraged the APIs of collaboration tools to extract data efficiently and reliably.
- Developed an ingestion framework using Python Flask to process incoming data and scheduled scraping weekly based on Python Flask.
- Designed and implemented MySQL database and Solr core schema structures to facilitate efficient data ingestion and retrieval.
- Created a system using self hosted **LLM (Llama 3) using TGI inference** docker based for generating reusable Q&A from threads, classifying them into categories such as Technical Issues, Individual User Issues, System Features, and Process Guidance to streamline retrieval and reduce incorrect answers.
- Used **multi-threading** to enhance performance and ensure efficient batch data processing while managing API rate limits.

Web Crawling and Multimodal Data Parsing Solutions

Roles and Responsibilities: -

- **Web Crawling:** Developed a Python web crawler with BeautifulSoup, reducing client onboarding time from days to hours.
- **PDF Parsing:** Implemented structured data extraction from PDFs using **PyMuPDF** and **GROBID**.
- **Image Parsing:** Extracted text from images using PyTesseract for information retrieval.
- **Video Parsing:** Fine-tuned **AWS Transcribe** and integrated **OpenAI Whisper** for accurate video transcriptions.
- **Data Integration:** Applied **semantic chunking** using LLM to unstructured data from PDFs, videos, and web crawled content, improving retrieval accuracy by **5-7%**.
- Created and deployed production-ready code using Flask, Gunicorn, and relevant Python libraries, utilizing Docker and CI/CD pipelines on **AWS**.

Performance Optimization of Re-ranker Model

Roles and Responsibilities: -

- Optimized the re-ranker model using **AWS Neuron** architecture and **Inferentia** instances, achieving up to **1.8x** higher throughput, **60%** lower latency, and a **37%** reduction in operational costs per inference.
- Implemented batch inference strategies and multiprocessing techniques to enhance data processing efficiency and scalability.
- Created a deployment pipeline using Neuron Docker images and CI/CD practices.

Automated Question Generation from Document Segments

Developed an automated question generation framework using the **Hyde technique**, leveraging **LLAMA3** to enhance the generation of relevant questions from text chunks, significantly improving information retrieval.

Roles and Responsibilities: -

- Utilized a technology stack of Python Flask, Gunicorn, LLAMA3, **Langchain**, Docker, and CI/CD for efficient deployment.
- Self-hosted **LLAMA3** on AWS GPU instances using Docker and **TGI inference** for optimal scalability and performance.
- Designed an automated weekly process using a Flask-based scheduler to parse historical PDFs, generate questions, and index them into Solr and MySQL, leveraging multi-threading to reduce processing time.

Data Scientist

Tech Mahindra Pune | From May 2022 to May 2023

Roles and Responsibilities: -

- **Image Analytics Workbench:** Developed a workbench with data augmentation, crowd-sourced data annotation, and Transfer learning using **YOLOv5**, **Faster R-CNN**, **ResNet**, and **VGG**. Responsibilities included backend design, REST API development with **Django**, and deployment via **Docker** and **AWS**.
- **Generative AI Workbench:** Showcased **OpenAI GPT-3** for applications like question answering and code generation using zero and few-shot learning. Led backend design, REST API development with **Flask**, and deployment with Docker on AWS.
- **AIOps Platform:** Created a platform for metric data anomaly detection and log clustering. Responsibilities encompassed metric forecasting with **Prophet**, log clustering using **Logpai**, data engineering pipelines with **Airflow**, and model deployment on Docker and AWS.

Data Science Intern

ABI Health | From April 2019 to May 2019

BRCA Test Prediction (Oncology)

BRCA Test used for detection of Breast and Ovarian cancer. In this project data used is insurance claim data of patients. Using patient journey (i.e. claim data) predicting whether he or she requires undergone to BRCA Test. Stacked **GRU** Model architecture used for prediction of BRCA Test.

Malignant Nodules Classification and Detection from X-Ray

Classifying the malignant nodules in Chest X-ray images as abnormal using state of the art **CNNs** like **DenseNet-121**, **ResNext50**, Transfer-Learning by **fast.ai** libraries to help Radiologists (**89% AUC** is achieved in classification which is the best in industry).

Data Scientist

Shyena Tech Yarns | From Sep 2018 to March 2019

Roles and Responsibilities: -

- Developed dashboards for real-time sensor data monitoring, including data collection, cleaning, and clustering analysis, using **Grafana**, **PostgreSQL**, Python, Docker, and **AWS EC2**.
- Implemented inventory demand forecasting with time series analysis using models like **S-ARIMA**, **Holt-Winters**, **LSTM**, and **XGBoost** to optimize inventory levels.
- Worked on a computer vision project for defect detection using **YOLO** models; fine-tuned the model on a GPU server and deployed it on AWS.

PERSONAL PROJECTS

Rakuten Supervised Recipe Food Image Classification Competition

Participated in a competition to identify food items from recipe images using **CNN** with transfer learning; achieved **59th** rank.

Bottle Counting in manufacturing Industry using computer vision approach

Developed a computer vision model using **Faster R-CNN** to count daily bottle production, implementing a counting algorithm for accuracy.

Quora Question pair Similarity

Built a model to assess the similarity of questions on Quora using text processing and machine learning algorithms like **SVM**, **Logistic Regression**, and **XGBoost**.

EDUCATION

M.Tech in Electronics (2016-2018) From VJTI Mumbai with CGPA 8.52.

B.E in Electronics and Telecommunication From SKNCOE, Pune with 63.61%